JACOB PILAND

Empirical AI Safety & Machine Learning Research Engineer
Alignment-Relevant ML | Interpretability | Robustness | Calibration | Multimodal Evaluation
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PROFESSIONAL SUMMARY

Ph.D. machine learning researcher-engineer whose work bridges trustworthy AI and empirical AI safety. I develop measurable methods for detecting and reducing mismatches between model behavior, model explanations, and human intent. My research examines misleading explanations, unreliable confidence, brittle evidence use, distribution shift, adversarial pressure, and the ways structured human expertise can improve multimodal model decisions. Across these projects, I build rigorous experimental systems that make models more reliable and human and AI interactions safer, more interpretable, and more accountable.

CORE COMPETENCIES & TECHNICAL SKILLS

Alignment-Relevant ML / Empirical Safety	Model behavior evaluation; calibration; adversarial robustness; OOD/open-set evaluation; safeguards evaluation; stress-testing; human-guided oversight; interpretable ML
Interpretability & Model Steering	GradCAM/CAM methods; saliency-guided training; contrastive explanations; explanation robustness; entropy-guided model focus; decision-aligned attribution
Research Engineering & ML Methods	Python; PyTorch; experiment design; reproducible pipelines; statistical benchmarking; multi-run evaluation; model training; code/model release; technical writing; energy-based models; multimodal vision-language models; prompt-based evaluation

EDUCATION

Ph.D., Computer Science | University of Notre Dame du Lac	**May 2026**
M.S., Computer Science | University of Notre Dame du Lac	**May 2021**
B.S., Computer Science; B.S., Biochemistry | Utah State University	**May 2018**

RESEARCH EXPERIENCE

Doctoral Candidate / Researcher | University of Notre Dame | *South Bend, IN* **2021-2026**

- Designed and executed a 16-condition study across a frontier multimodal model and a locally deployed 90B vision-language model, showing how structured instructions, human-provided expertise, and machine-expanded guidance can steer model behavior in a specialized high-stakes task.
- Introduced Salience-Hoax Activation Maps, an adversarial interpretability benchmark in which models preserve task performance while producing systematically misleading explanations, enabling controlled evaluation of explanation faithfulness.
- Developed DiffGradCAM and DiffGradCAM++, lightweight interpretability methods designed to better reflect the evidence underlying a model's decision and remain reliable when standard explanations are adversarially manipulated.
- Developed NG-EBM, a non-generative energy-based training method that better aligns model confidence with empirical uncertainty, strengthens out-of-distribution and adversarial behavior, and reduces training cost sixfold by eliminating sampling-based generation.
- Built reproducible, multi-run evaluation pipelines spanning calibration, out-of-distribution behavior, adversarial robustness, open-set generalization, explanation fidelity, and multimodal model behavior, including experiments across 700,000-image benchmarks; released source code, model weights, and training configurations.
- Extended decision-aligned interpretability into model training by supervising both supporting and opposing evidence, improving generalization across three distinct high-stakes visual domains.
- Developed DROID and REBIS to study how training-time control of model evidence interacts with calibration, open-set generalization, and adversarial robustness, demonstrating the value of

optimizing multiple reliability properties rather than accuracy alone.

ADDITIONAL RESEARCH, ENGINEERING & COLLABORATION EXPERIENCE

Research Assistant | University of Notre Dame | *South Bend, IN* **2018-2021**

- Developed network-science methods for protein-structure analysis without ground-truth labels, using interpretable proxy measures and translating among collaborators in biochemistry, statistics, and computer science.

Research Assistant | Utah State University | *Logan, UT* **2014-2018**

- Conducted large-scale statistical research on how computer-based scaffolding shapes human reasoning and learning across U.S. and international cohorts; contributed to a funded proposal for machine-learning-enhanced adaptive instruction.

Programming Consultant | NASA Earth Science (Armstrong / UC Irvine) | *Irvine, CA* **Summer 2018**

- Advised software development across 28 interdisciplinary Earth-science projects in Python, C++, MATLAB, and R, and delivered weekly programming instruction to improve research-code reliability, debugging practices, and scientific-computing workflows.

Software Engineering Intern | Hewlett Packard Enterprise | *Boise, ID* **Summer 2017**

- Improved the reliability and maintainability of a systems-level C codebase by increasing power-failure test coverage fivefold, reducing runtime by 30%, and substantially simplifying legacy driver code.

Research Intern | NASA Earth Science (Armstrong / UC Irvine) | *Irvine, CA* **Summer 2015**

- Modeled atmospheric ozone production from airborne sensor data, developing mathematical measures of reaction dynamics and operating laser-absorption instrumentation aboard a NASA DC-8.

Selected Publications

- **Piland, J.**, Dowling, B., Sweet, C., Czajka, A. (2026). Generalist Multimodal LLMs Gain Biometric Expertise via Human Salience. IEEE Access.
- **Piland, J.**, Sweet, C., Czajka, A. (2026). DiffGradCAM: A Class Activation Map Using the Full Model Decision to Solve Unaddressed Adversarial Attacks. Findings of CVPR.
- **Piland, J.**, Sweet, C., Czajka, A. (2023). Non-Generative, Energy-Based Models. IJCNN.
- **Piland, J.**, Sweet, C., Czajka, A. (2025). Multi-Level Entropy Guidance Improves Model Performance in Synthetic Face Detection. ICIP.
- **Piland, J.**, Sweet, C., Czajka, A. (Under review). Divisive Decisions: Utilizing Both CAMs in Salience-Based Training to Improve Generalization in Binary Classification Tasks. Submitted to IJCB.
- **Piland, J.**, Sweet, C., Czajka, A. (2023). Model Focus Improves Performance of Deep Learning-Based Synthetic Face Detectors. IEEE Access.

Leadership and Service

Nonprofit Regional Leader | Nonprofit Organization | *Tokyo, Japan* **2011-2013**

- Led and supported approximately 40 multinational volunteers; redesigned English-teaching materials and coordinated instruction, logistics, community service, and individual support.

Finance Lead | Utah State University | *Logan, UT* **2016-2017**

- Managed accounts totaling more than $70,000 for over 40 individuals, maintaining accurate allocation, reporting, and financial stewardship.

Financial Judge | Northern Indiana Regional Science & Engineering Fair | *South Bend, IN* **2020-2025**

- Evaluated student research for methodological rigor, scientific validity, communication, and potential impact; served on the committee responsible for allocating financial awards.